

AXO

GATE 1 — PROBLEM & PROOF

AI SmartBus

Predictive, Accessible & Crowd-Aware Public Transport

Item	Details
Team	AXO
Domain	Public bus transport • accessibility • mobility
Primary user	Daily/frequent bus passengers on a selected Sri Lankan corridor, especially where visibility or waiting uncertainty is a problem
Core problem	Passengers may not know which bus is approaching, when it will arrive, or how crowded it may be before boarding
MVP	Predicted ETA • predicted crowding • bus/route identification • voice/haptic alerts
Validation	Initial conversations with multiple bus passengers + review of official transport data and existing transit services

Prepared by Team AXO • September 2026

01. Problem & Domain

In Sri Lanka's public bus transport domain, passengers can face uncertainty while waiting for a bus: which bus is approaching, when it will reach their stop, and whether there will be usable space when it arrives. This becomes harder in rain, fog, darkness, heavy traffic or remote areas where an approaching bus may be difficult to identify. The issue is also relevant to passengers who cannot depend on visual information. Existing digital services show that live tracking and journey information are already useful, but a passenger still needs a simple decision-support layer before boarding: an early identification of the bus, a stop-specific arrival prediction and an indication of expected crowding. Our project therefore focuses on public bus mobility, passenger information and accessibility, with an initial corridor-level prototype.

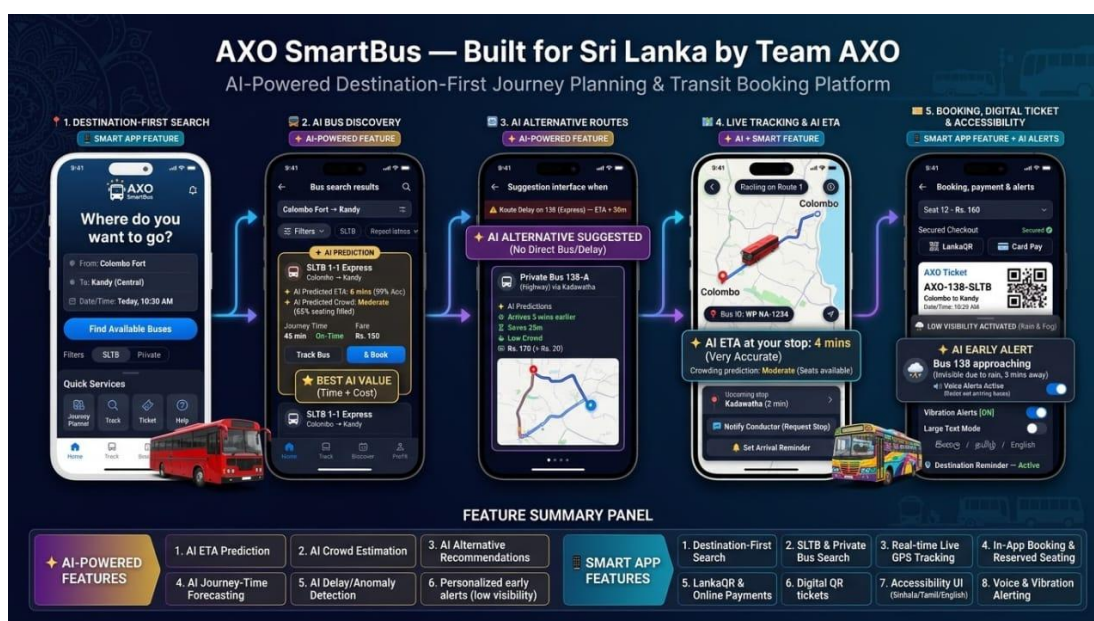
02. Exactly Who Has the Problem

Our primary user is a daily or frequent bus passenger waiting at a known stop on a selected route/corridor who needs to decide whether to continue waiting, board the approaching bus, or choose another option.

- Daily commuters who depend on buses for regular travel.
- Passengers travelling on routes affected by rain, fog, darkness or poor visibility.
- Passengers who want to know likely crowding before the bus reaches their stop.
- Visually impaired or other passengers who benefit from voice and vibration instead of relying only on visual information.
- Passengers in remote or low-connectivity areas, where lightweight notification channels may be useful.

03. Proof the Problem Is Real

Before defining the final solution, Team AXO spoke with multiple bus passengers to understand their day-to-day experience of waiting for and boarding buses. We asked about uncertainty over arrival time, difficulty identifying the correct approaching bus, crowding before boarding, and whether advance voice, vibration or visual alerts would be useful. These conversations gave us initial qualitative evidence that passenger information before boarding is worth investigating. We are treating this as early validation, not as a statistically representative survey, and we are not inventing participant counts, percentages or quotations. In addition to speaking with passengers, we reviewed official Sri Lankan transport statistics and current public-transport services. The Department of Census and Statistics reports 2,052,018 thousand omnibus passengers and 946,416 thousand Sri Lanka Transport Board passengers in 2024, showing the scale of bus use. The National Transport Commission documents MyBus-SL and GPS-based bus information, while current services such as WayToGo and LMT-GO provide live tracking and other passenger information. This research confirms that the transport-information problem is relevant, while also showing that our project must differentiate itself from existing tracking services. The next validation step is to record structured interview notes and compare passenger needs across the selected corridor.



04. Market Size

The Department of Census and Statistics reports 2024 passenger figures of 2,052,018 thousand for omnibus services and 946,416 thousand for the Sri Lanka Transport Board. Together these give approximately 2.998 billion reported passenger units for the two categories.

Level	Estimate	Assumption / meaning
TAM	~2.998 billion reported passenger units/year	Broad planning measure based on 2024 reported bus passenger figures.
SAM	~299.8 million/year	10% planning assumption for an initial serviceable segment.
SOM	~3.0 million/year	1% of SAM used as an early adoption planning target.

Important clarification: these figures are an activity/volume measure reported by the source, not unique passengers and not revenue. They should not be presented as an exact count of unique people or as a monetary market size. The SAM and SOM percentages are AXO planning assumptions, not official forecasts.

05. Alternatives & Competitors

The competitive baseline is already active in Sri Lanka. Therefore, AXO does not claim that live tracking, ETA, occupancy or accessibility are individually new.

Alternative / competitor	Documented capability	Problem-space difference AXO will test
NTC MyBus-SL	GPS-based bus location, schedules/fares, seat reservation, complaints and inquiries.	Test a predictive layer combining stop-specific ETA, expected crowding and early passenger alerts.
WayToGo	Live bus tracking, Smart ETAs for a specific halt and 2,000+ offline schedules.	ETA and tracking are already available; AXO will test whether crowd-at-arrival prediction and accessibility-focused alerts add value.
LMT-GO	Real-time tracking, arrival alerts, occupancy, journey planning, accessibility information, offline mode and multilingual support.	AXO focuses initially on a selected corridor and tests predictive crowding plus early bus identification and voice/haptic delivery.
Central Province Live Bus Tracking	Live bus positions by route/direction where GPS data is transmitting.	AXO moves from current location toward prediction at the passenger's selected stop.

Competitive positioning: The proposed contribution is the combination of arrival and crowding prediction, early bus/route identification and multi-channel accessibility alerts, validated against real passenger needs and measured prototype performance. This is a positioning hypothesis, not a claim that existing services lack value.

06. Proposed Solution

AI SmartBus will provide a passenger with the identity of an approaching bus, its predicted arrival time at the selected stop, and an expected crowd level before boarding. A physical bus prototype will send GPS and passenger-count data to a backend. The prediction layer will learn travel-time and passenger-flow patterns and produce ETA and crowd estimates. The passenger interface will present the information visually and can issue voice and vibration alerts, making the system useful when a bus is difficult to see or when the passenger cannot rely on visual information. For the initial buildathon MVP, AXO will focus on one demonstration corridor and four outputs: bus/route identification, predicted ETA, predicted crowding and accessible alerts. Future extensions can include SMS/USSD or stop-side audio/display for users without smartphones, driver/conductor communication for verified stop requests, and adaptation of the same architecture to trains and other public transport.

07. Why Does This Need AI?

GPS can show where a bus is now, but ETA and crowding depend on changing travel and passenger-flow patterns. AI can learn from GPS position, speed, route, time/day, historical travel time and occupancy/boarding patterns to predict what will happen at a selected stop. The prototype will compare ML predictions against simple baselines such as timetable-based ETA and current-speed GPS estimation; AI will be retained only if it provides measurable improvement.

08. Technology Stack, Three-Week Plan & Team

Technology stack:

- Prototype: ESP32 or Raspberry Pi, GPS module and passenger-counting sensor(s); optional camera for later occupancy estimation.
- Backend/API: Python + FastAPI; PostgreSQL/PostGIS for route and location data.
- AI/ML: Python with XGBoost or LightGBM for ETA/crowd prediction; computer vision only if needed.
- Communication: MQTT/WebSocket; SMS/USSD as a later fallback.
- Frontend: Flutter or React Native.

Three-week plan:

Week	Work	Output
Week 1	Complete structured passenger validation, select corridor, confirm data sources and prepare telemetry.	Interview notes + data plan + baseline design
Week 2	Build the minimum data pipeline and ETA/crowd baselines; train initial ML models.	Prediction results + baseline comparison
Week 3	Connect passenger interface, voice/haptic alerts and end-to-end prototype testing.	Working prototype + measured demo results

Team: AXO. The confirmed member names should match the roster registered on the buildathon platform; no names have been invented in this report.

Supporting Evidence

No fabricated screenshots are included in this report. The primary validation is based on Team AXO's conversations with multiple passengers and documented external research. Interview notes, survey exports or photographs can be attached separately if available.

Sources

- Department of Census and Statistics — Statistical Pocket Book 2025, Transport and Communication. <https://www.statistics.gov.lk/Publication/PocketBook2025>
- Department of Census and Statistics — Statistical Abstract 2025, Chapter VII: Transport and Communication. <https://www.statistics.gov.lk/abstract2025/chapter7>
- National Transport Commission — Statistics Report 2023, including MyBus-SL. https://www.ntc.gov.lk/corporate/pdf/2023/statistics_Report/Stat_2023_EN.pdf
- National Transport Commission — Statistics Reports page. https://www.ntc.gov.lk/corporate/media_and_news/statistics.php
- WayToGo (Sri Lanka) — Google Play listing, updated August 16, 2026. <https://play.google.com/store/apps/details?id=com.hadoc.waytogo>
- LMT-GO: Lanka Metro Transit — Google Play listing, updated June 11, 2026. <https://play.google.com/store/apps/details?id=com.lankametrotransit.lmtgo>

Our app implement dashboard

